Using Closed-loop Learning as an evaluator of clustering quality

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1 Measuring the fluctuations of the centroids

We can take the KMeans algorithm as an example. The closed-loop learning (CLL) algorithm is iterative, with the following steps in each cycle:

- Apply KMeans clustering to the dataset, obtaining a hard labeling of the points.
- For each cluster, compute the mean and variance; this defines a mixture of Gaussians, one Gaussian per cluster.
- Generate M synthetic points from the Gaussian mixture defined above and add these points to the dataset.
- Repeat the process, with the dataset growing at each step.

For example this is what happens to a KMeans clustering of the Iris Dataset after a CLL of 200 steps with $M = 20$:

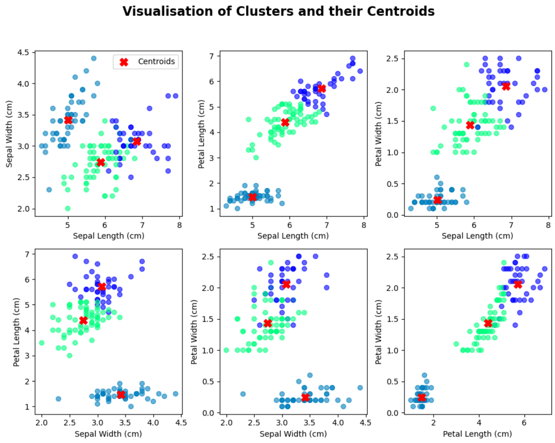


Figure 1: Before CLL

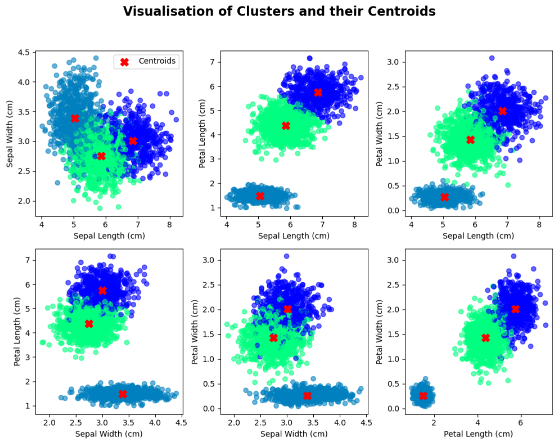
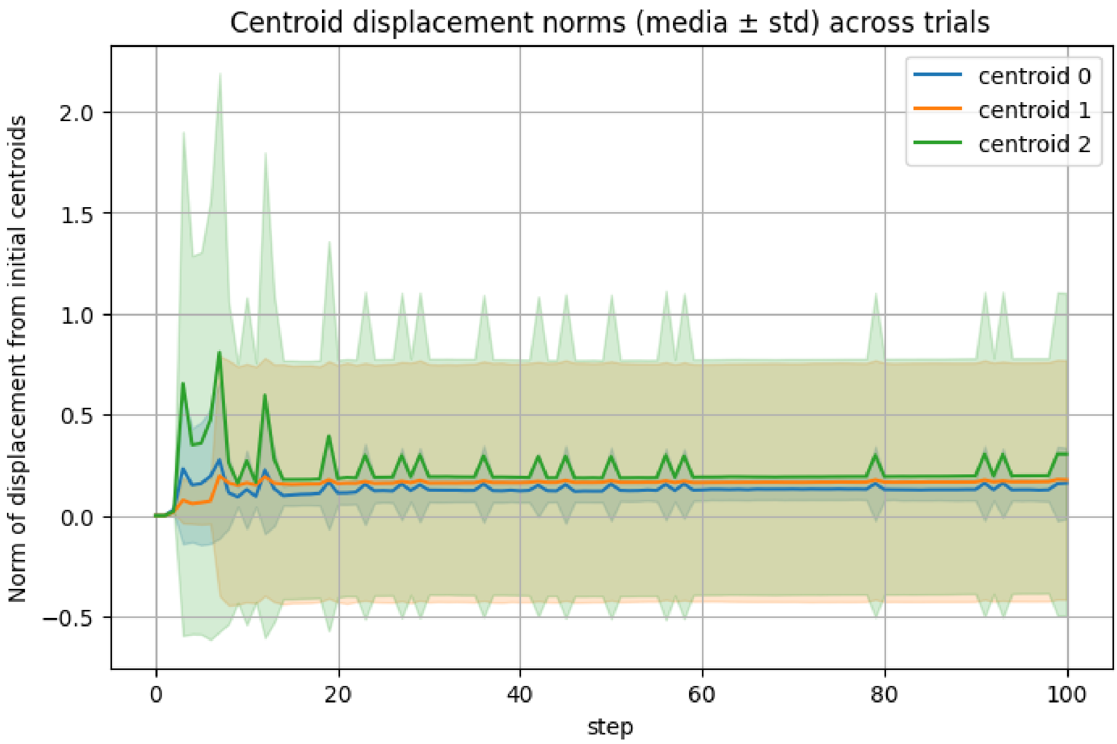


Figure 2: After CLL (200 steps, $M = 20$)

We expect that, after the CLL algorithm has converged, the cluster centers will have shifted slightly.

Assuming that CLL is a valid, convergent clustering procedure, we can assess the quality of the original method (here KMeans) by measuring the displacements and fluctuations of the centroids. We can plot the norm of the displacements of the centroids from initial conditions at each step of the algorithm:



2 Comparison with distance between clusters

The idea is that if we compare the displacements of the cluster δ after the CLL exection and the original distances D between the clusters we can evaluate the quality of the clusterisation.

We expect that:

- the smaller the ratio $\frac{\delta}{D}$ the better the clusterisation
- If the displacements δ are comparable with the original distances D, we consider it a poor clusterisation.

Given N clusters we can define the cluster quality Q as:

$$Q = \frac{\sum_{i=1}^{N} \frac{\delta_i}{D_i}}{N}$$

where D_i is the mean of the distances between cluster i and all the others, this is to say $D_i = \sum_{j=1}^{N} \frac{\|\overrightarrow{x_i} - \overrightarrow{x_j}\|}{N-1}$ where $\overrightarrow{x_i}$ is the centroid of cluster i

For example in the case of a KMeans clusterisation of the Iris Dataset we get:

$$Q = \frac{\frac{0.083}{3.39} + \frac{0.057}{4.17} + \frac{0.030}{2.57}}{3} = 0.017$$

While in the case of a Gaussian Mixtures clustering of the Iris dataset we get:

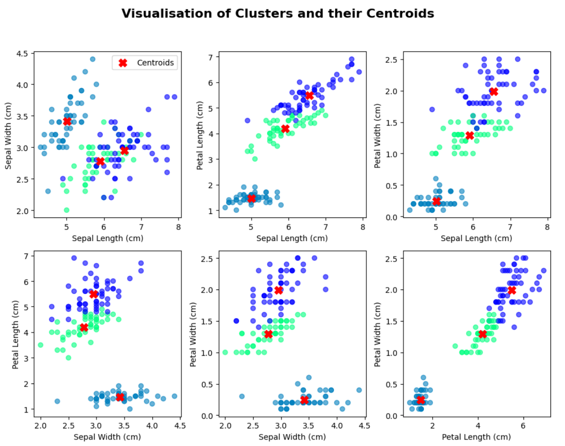


Figure 3: Gaussian Mixtures before CLL

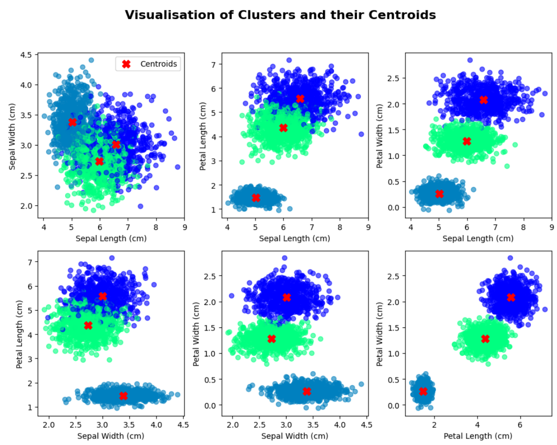


Figure 4: Gaussian Mistures after CLL (200 steps, $M = 20$)

We get these displacements of the centroids at each step:

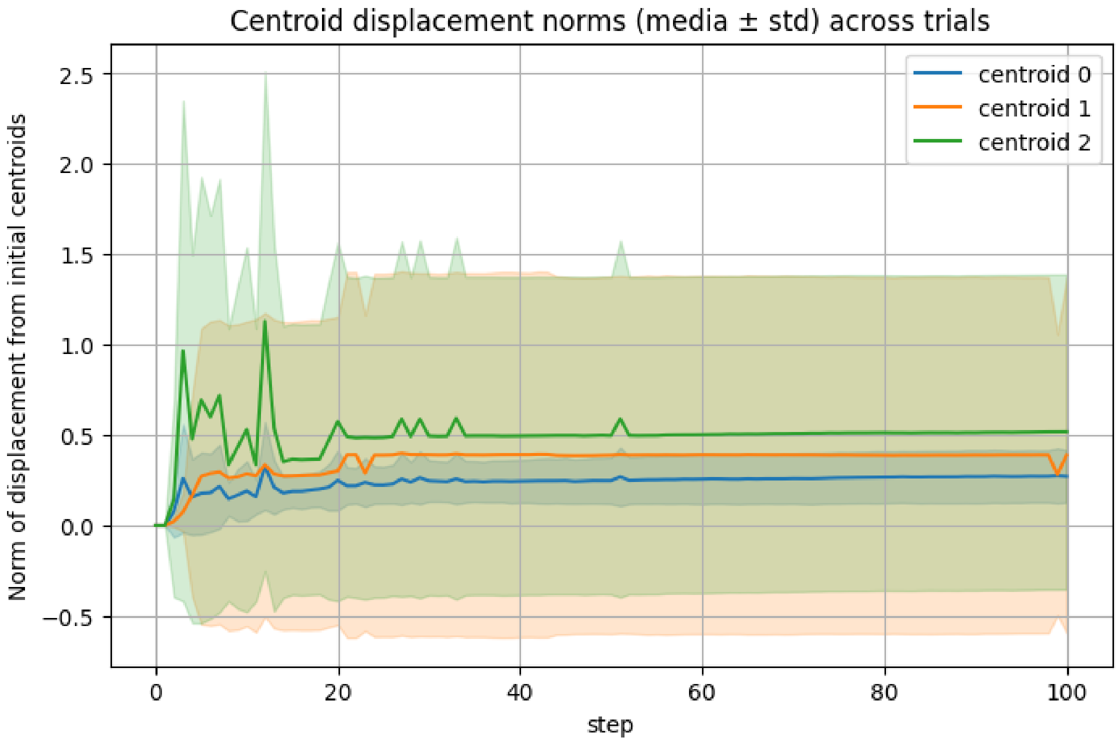


Figure 5: Displacements of centroids with Gaussian Mixtures on Iris database

and we get a quality of:

$$Q = \frac{\frac{0.131}{3.15} + \frac{0.044}{3.19} + \frac{0.202}{2.38}}{3} = 0.046$$

much worse than the quality of the KMeans clustering of Iris Database.